

What is it they call

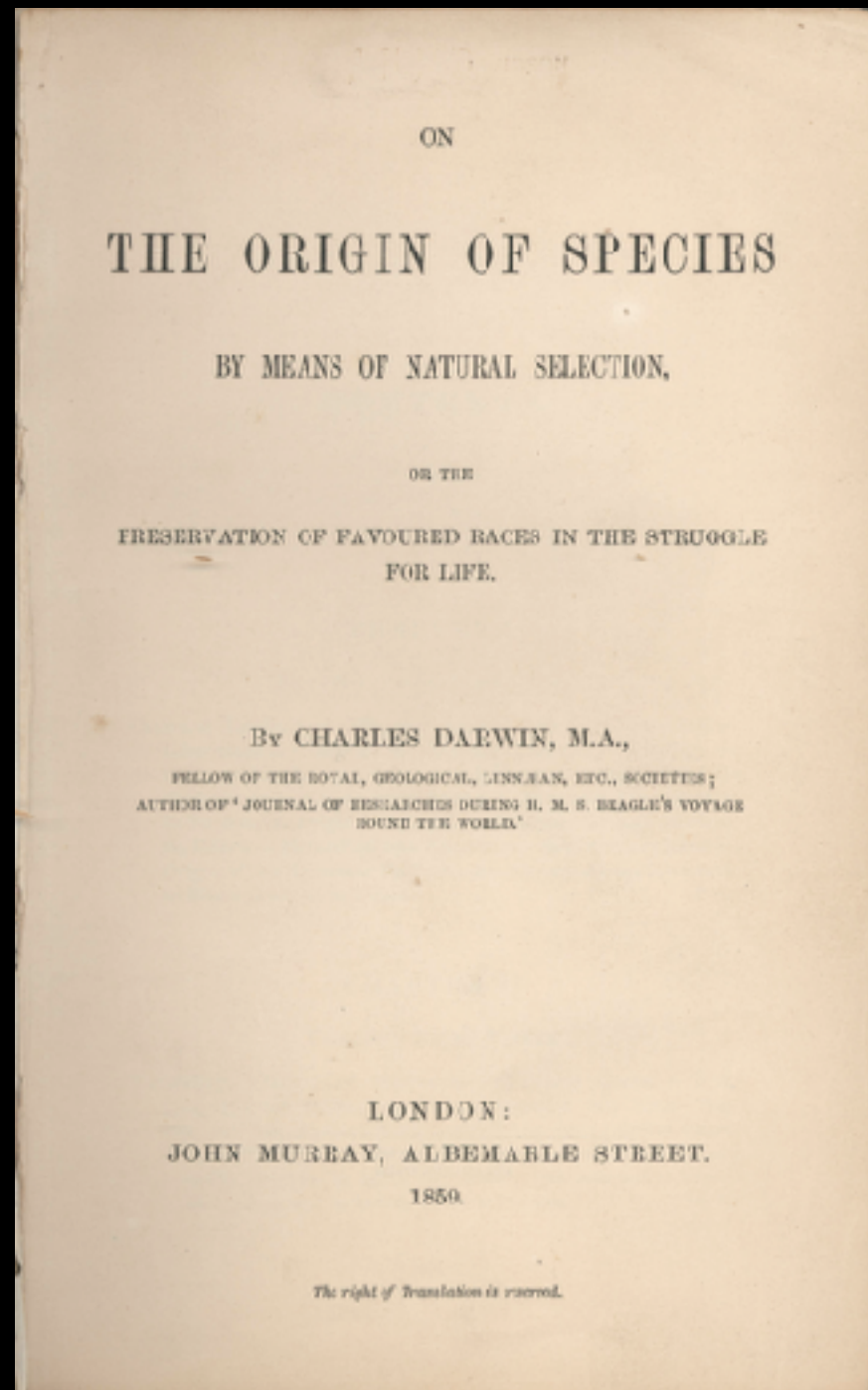
Lamarckism?

Ata Kalirad

School of Biological Sciences (IPM)

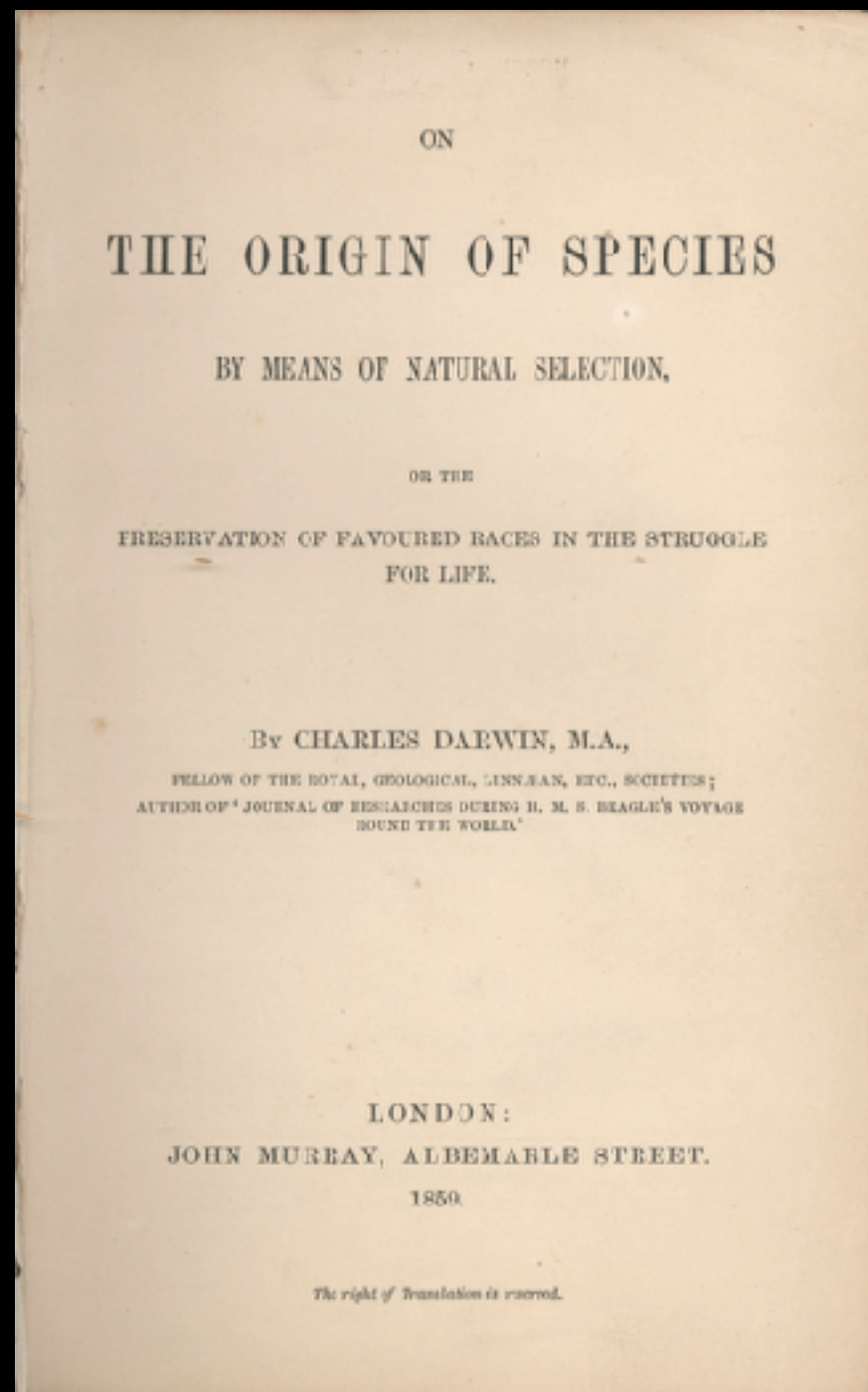
1 March, 2019

2009, an annus mirabilis

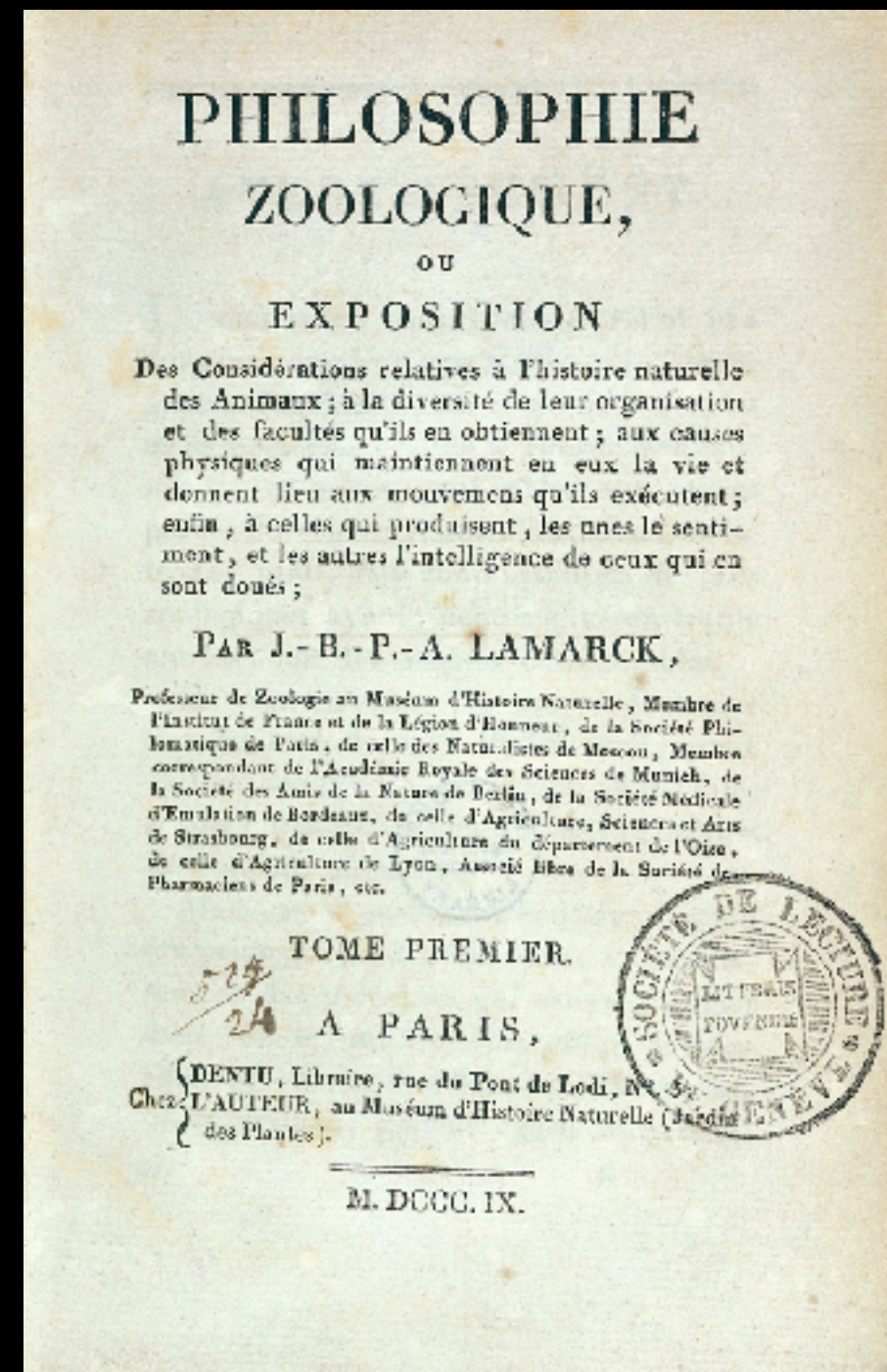


Published 1859

2009, an annus mirabilis

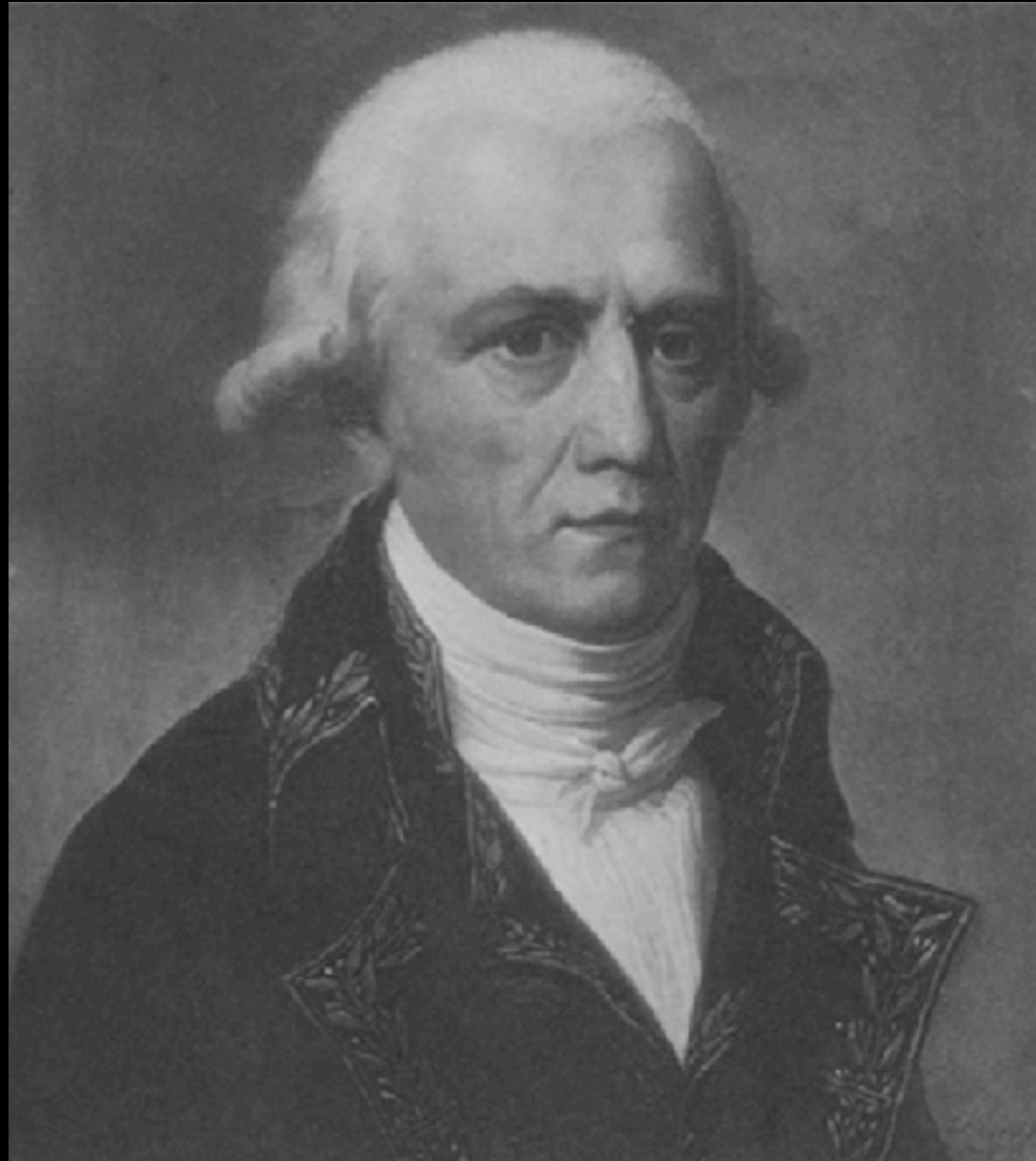


Published 1859



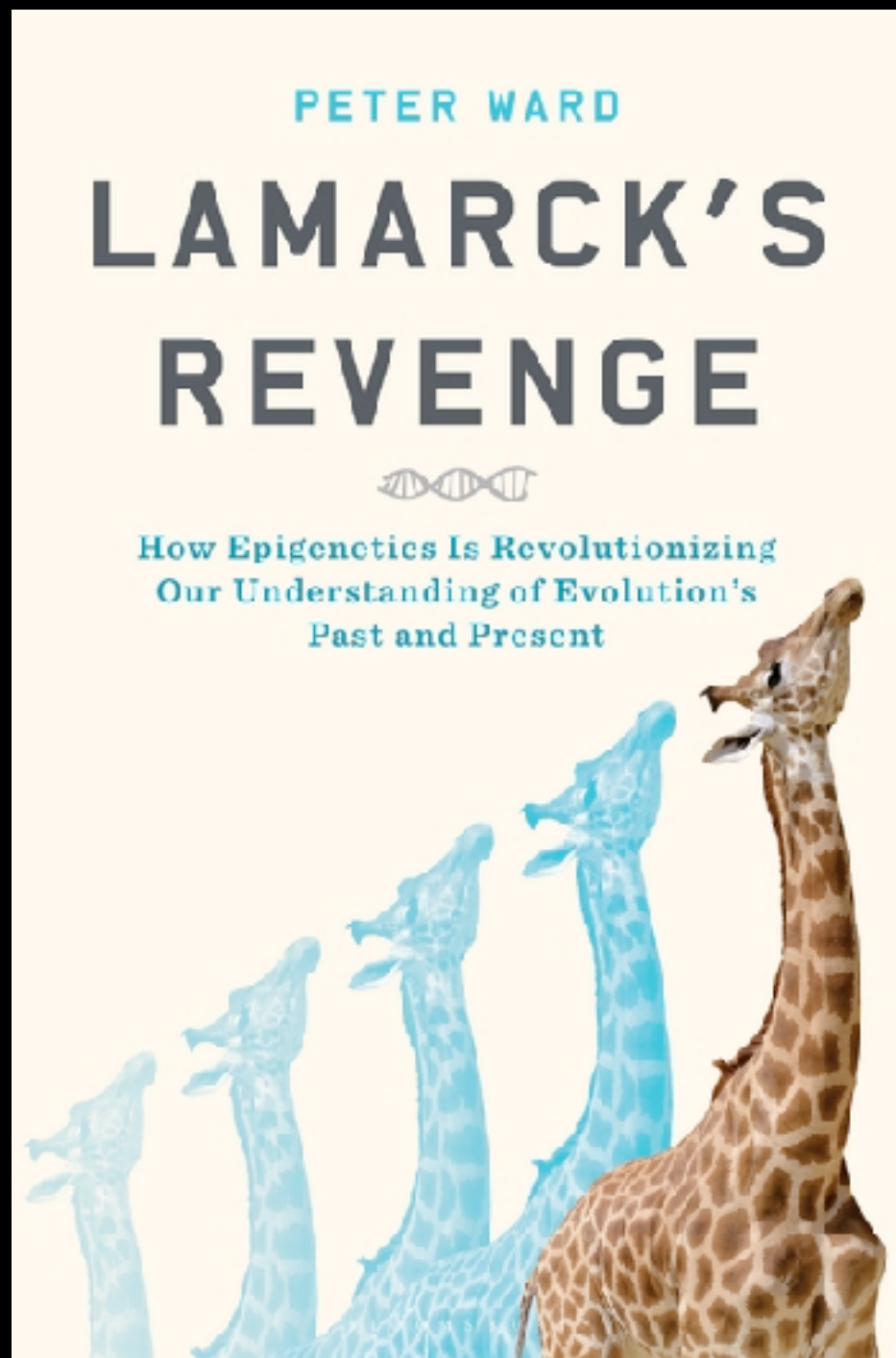
Published 1809

A revolution in Biology?



Jean-Baptiste Lamarck (1744-1829)

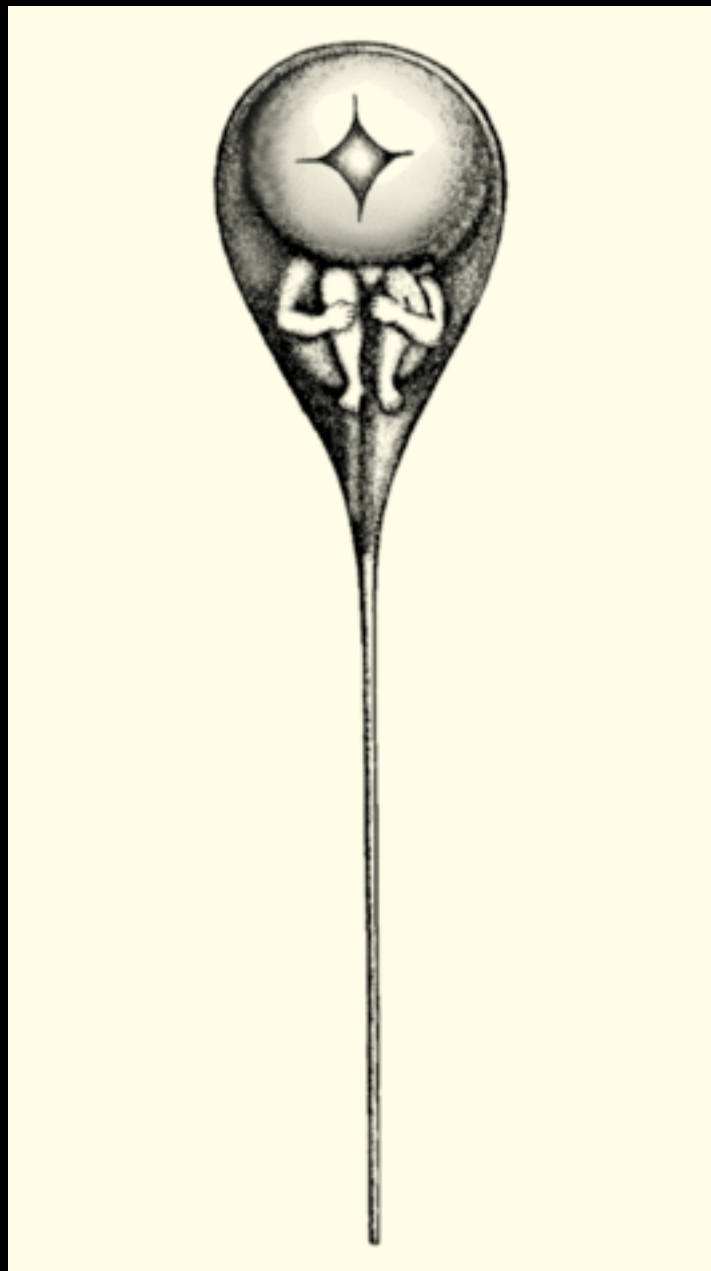
A revolution in Biology?



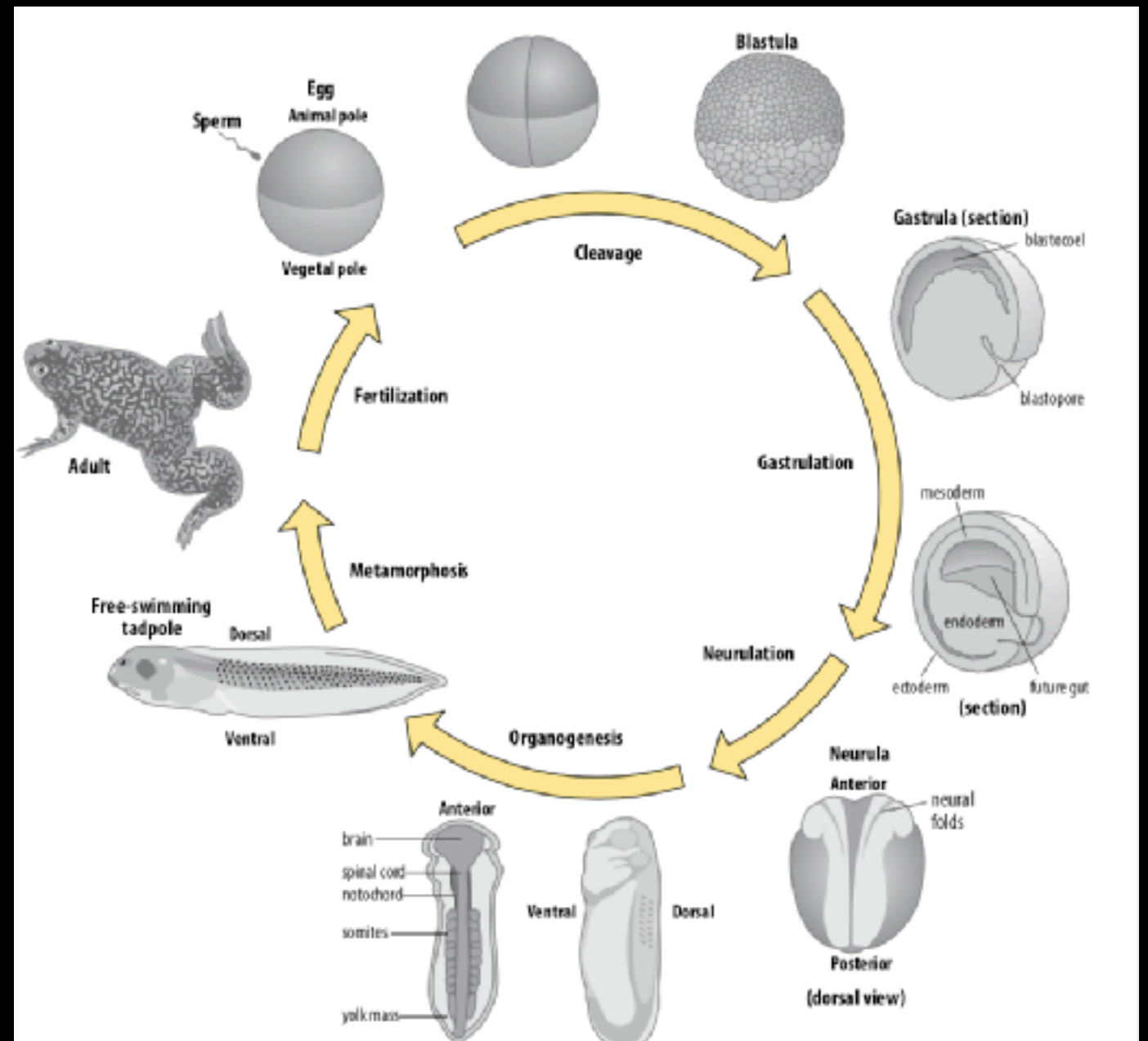
Published 2018

Today, a revival of Lamarckism is causing many biologists to admit that a major revolution in evolutionary theory might be at hand.

Epigenetics and Neo-Lamarckism



Preformationism



Epigenesis

Epigenetics and Neo-Lamarckism

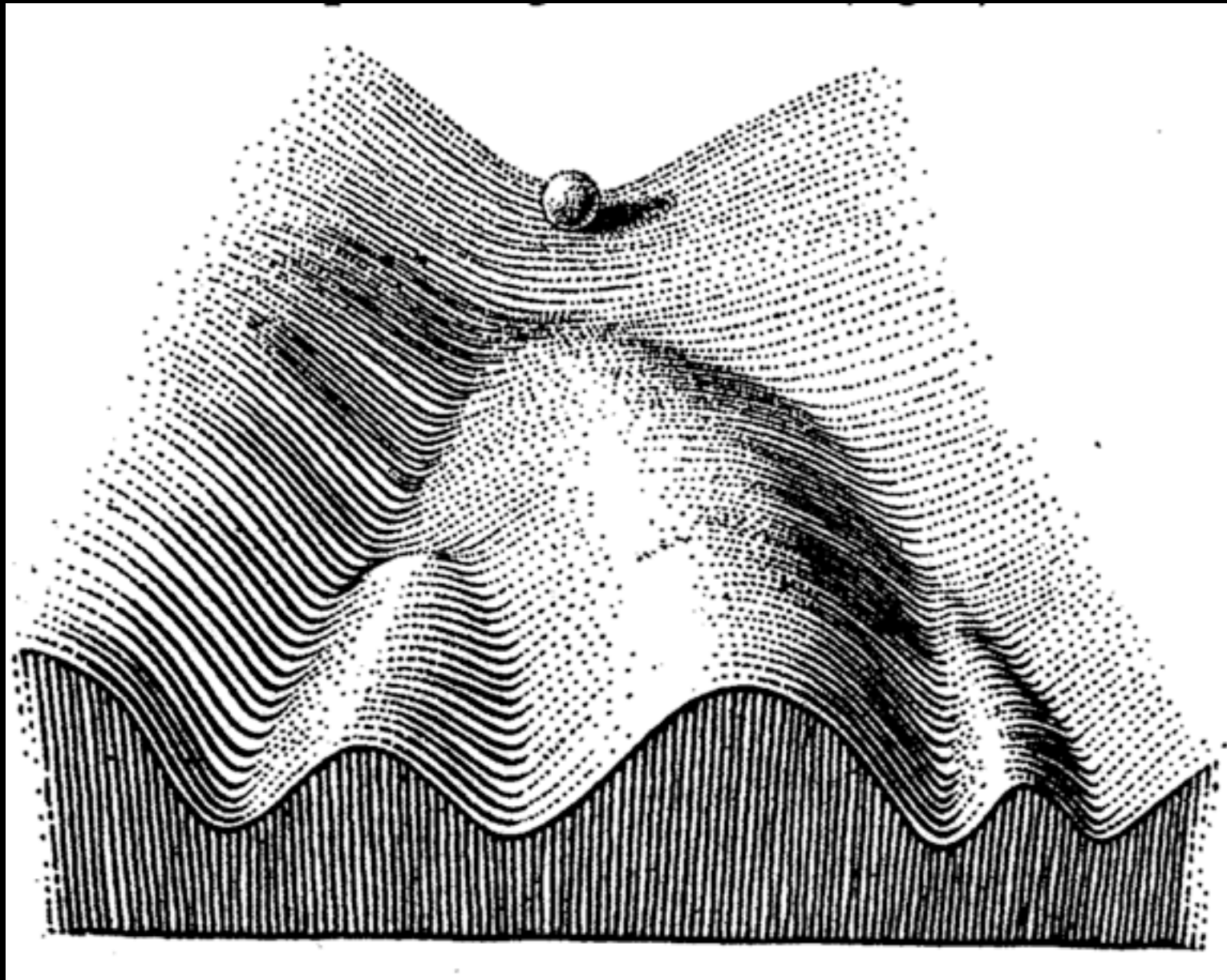


Conrad Waddington (1905-1975)

[Epigenetics] is the branch of biology that studies the causal interactions between genes and their products which bring the phenotype into being.

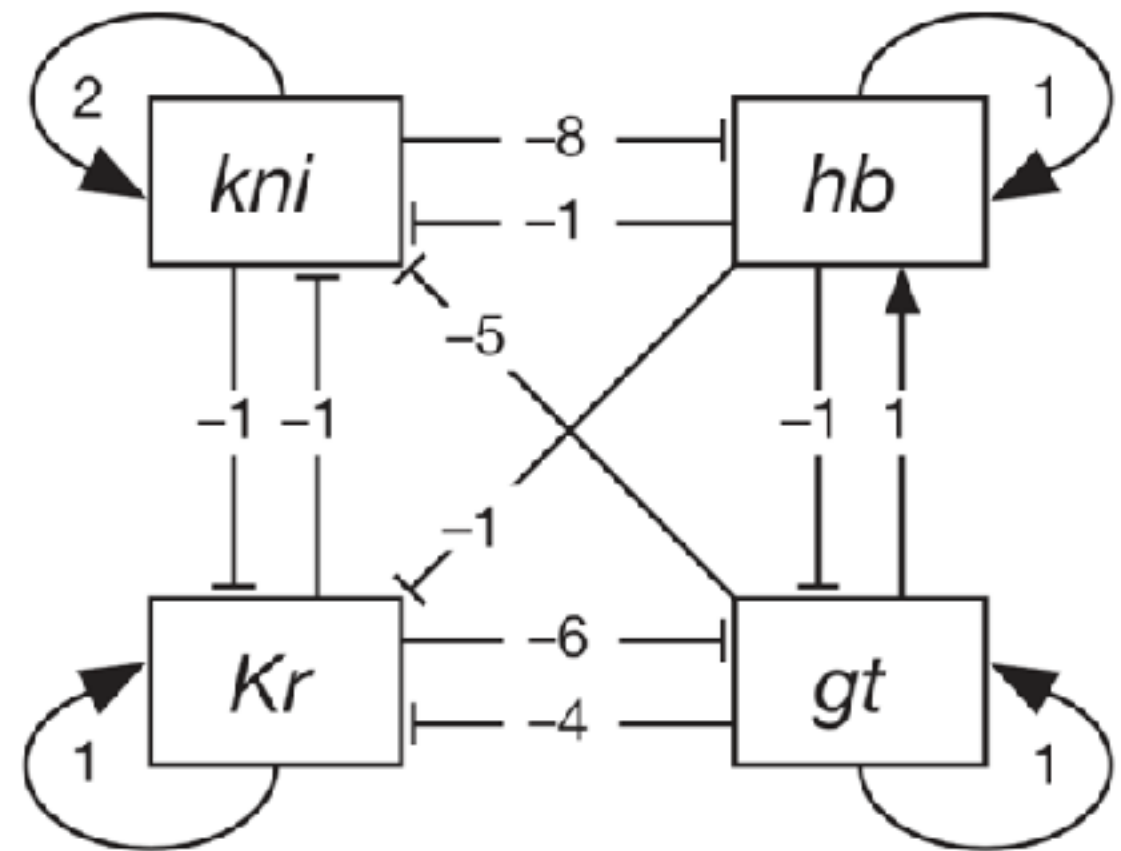
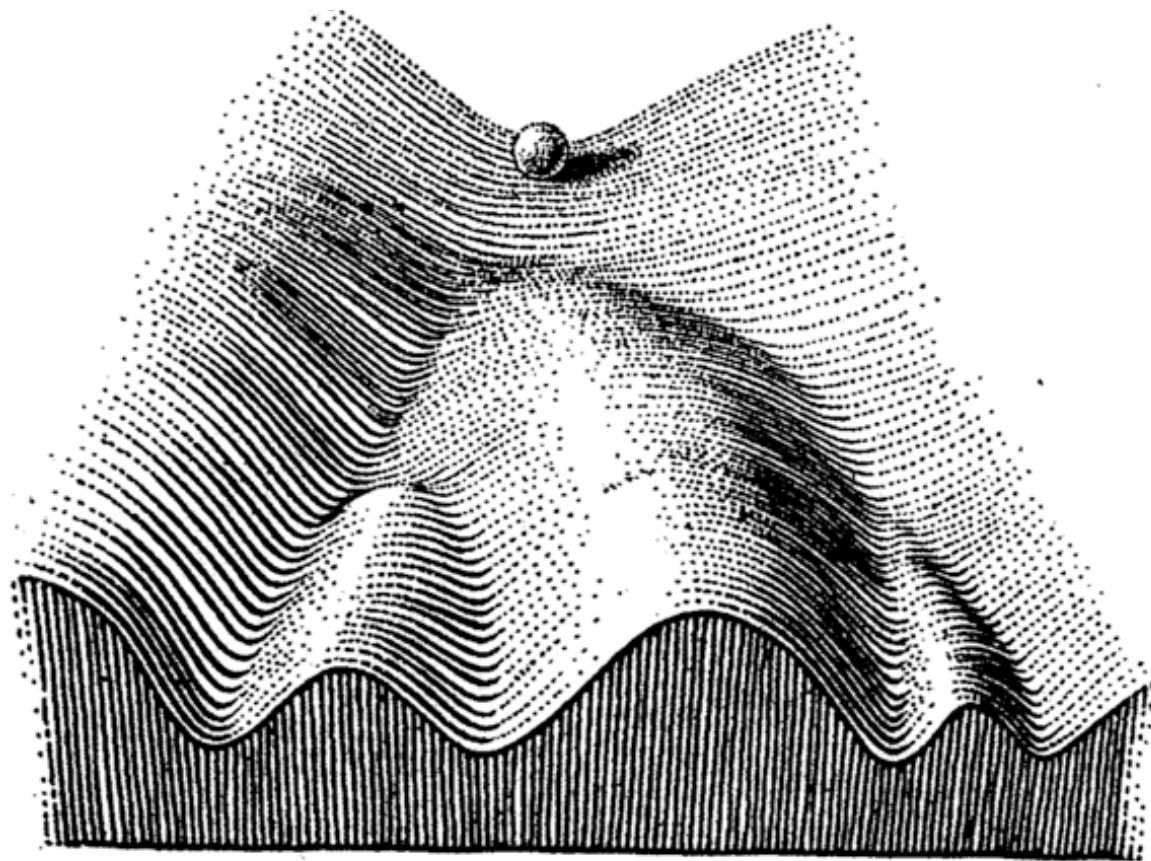
1942

Epigenetics and Neo-Lamarckism



Epigenetic landscape, from *the Strategy of the genes*, 1957

Epigenetics and Neo-Lamarckism



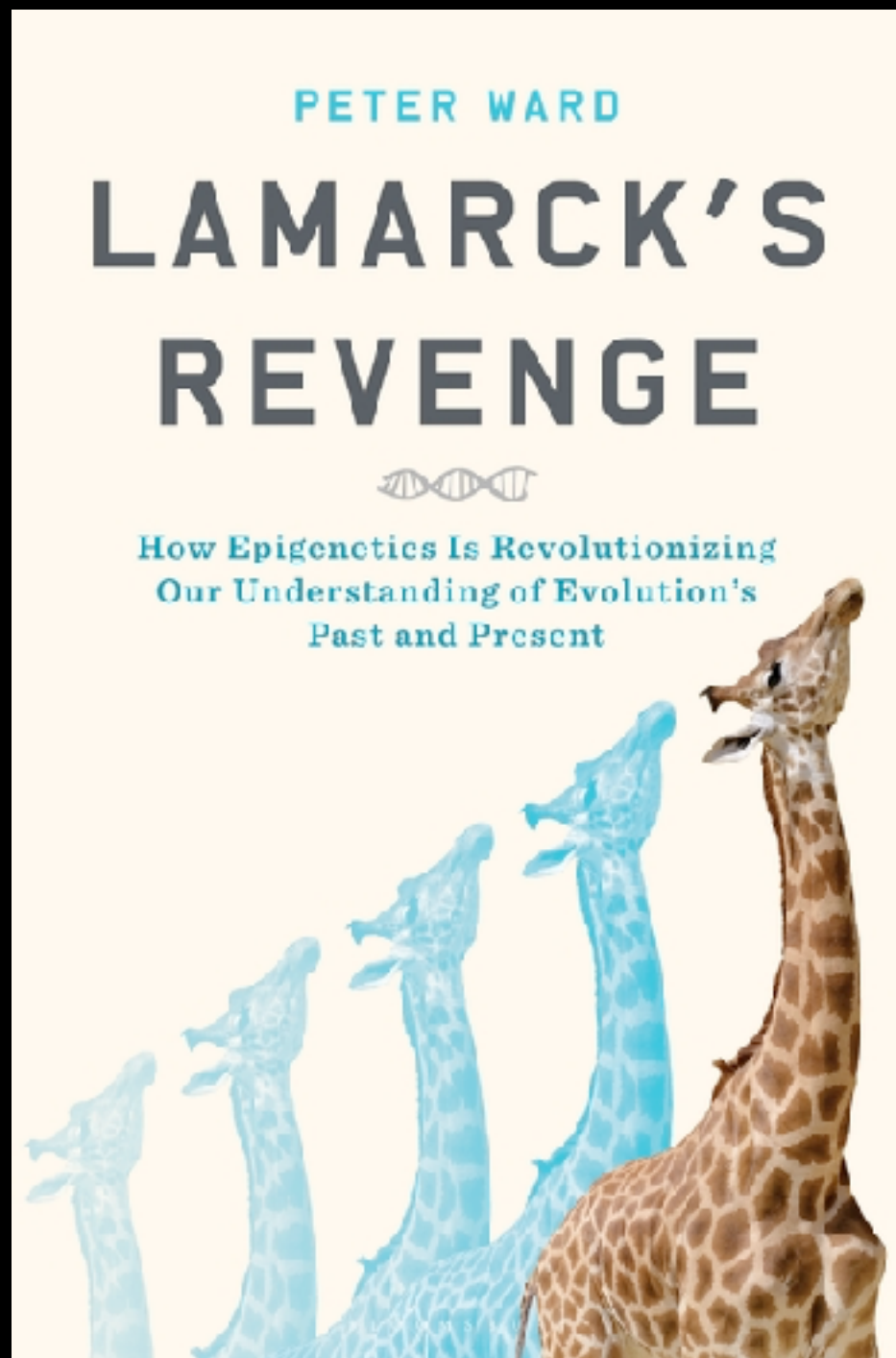
Azevedo et al., 2006

Epigenetics and Neo-Lamarckism

[Epigenetics] is the study of changes in gene function that are [...] heritable and that do not entail change in DNA sequence.

Wu and Morris (2001)

A revolution in Biology?



Published 2018

Evolutionary change in organisms [...] is driven by beneficial phenotypic [...] changes that were not randomly acquired but came about through an intersection of the organism with [...] environment—and that this intersection not only changed the life of the organism from that time onward, it was also heritable

Why neo-Lamarckism?

PHILOSOPHIE ZOOLOGIQUE, ou EXPOSITION

Des Considérations relatives à l'histoire naturelle des Animaux ; à la diversité de leur organisation et des facultés qu'ils en obtiennent ; aux causes physiques qui maintiennent en eux la vie et donnent lieu aux mouvemens qu'ils exécutent ; enfin , à celles qui produisent , les unes le sentiment , et les autres l'intelligence de ceux qui en sont doués ;

PAR J.-B.-P.-A. LAMARCK,

Professeur de Zoologie au Muséum d'Histoire Naturelle, Membre de l'Institut de France et de la Légion d'Honneur, de la Société Philomathique de Paris, de celle des Naturalistes de Moscou, Membre correspondant de l'Académie Royale des Sciences de Munich, de la Société des Amis de la Nature de Berlin, de la Société Médicale d'Emulation de Bordeaux, de celle d'Agriculture, Sciences et Arts de Strasbourg, de celle d'Agriculture du département de l'Oise, de celle d'Agriculture de Lyon, Associé libre de la Société des Pharmaciens de Paris, etc.

TOME PREMIER.

A PARIS,

(DENTU, Libraire, rue du Pont de Lodi, N° 3.
Chez L'AUTEUR, au Muséum d'Histoire Naturelle (Jardin
des Plantes).

M. DCCC. IX.



First Law

In every animal that has not reached the end of its development, the more frequent and sustained **use** of any organ will strengthen this organ little by little, develop it, enlarge it. [...] while the constant **disuse** of such an organ will insensibly weaken it

Why neo-Lamarckism?

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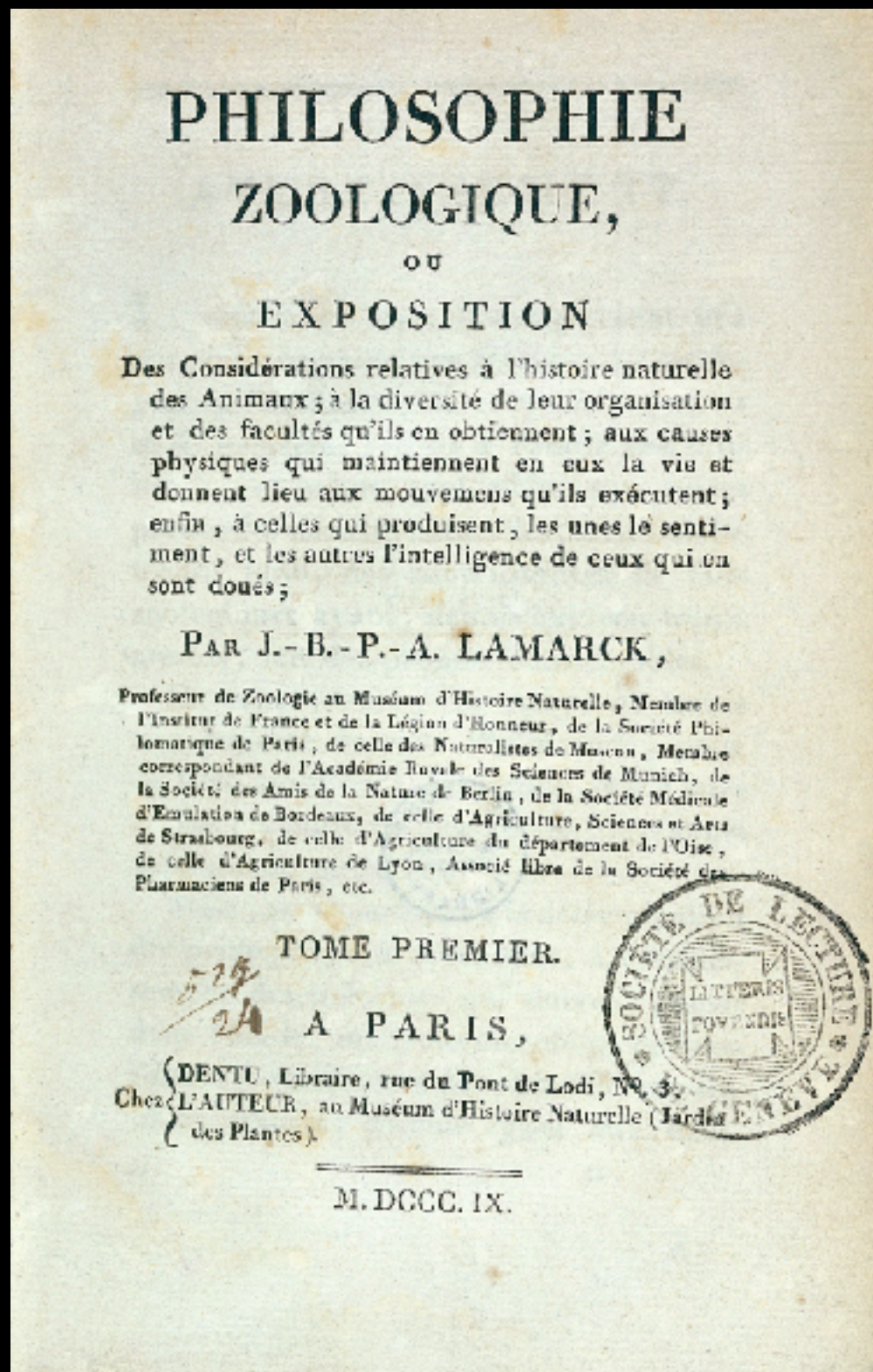
(DENTU, Libraire, rue du Pont de Lodi, N° 3.
Chez L'AUTEUR, au Muséum d'Histoire Naturelle (Jardin
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M. DCCC. IX.

Second Law

All that nature has caused individuals to gain or lose [...] by the influence of a predominant use or constant disuse of an organ or part, **is conserved through generation.**

Why neo-Lamarckism?



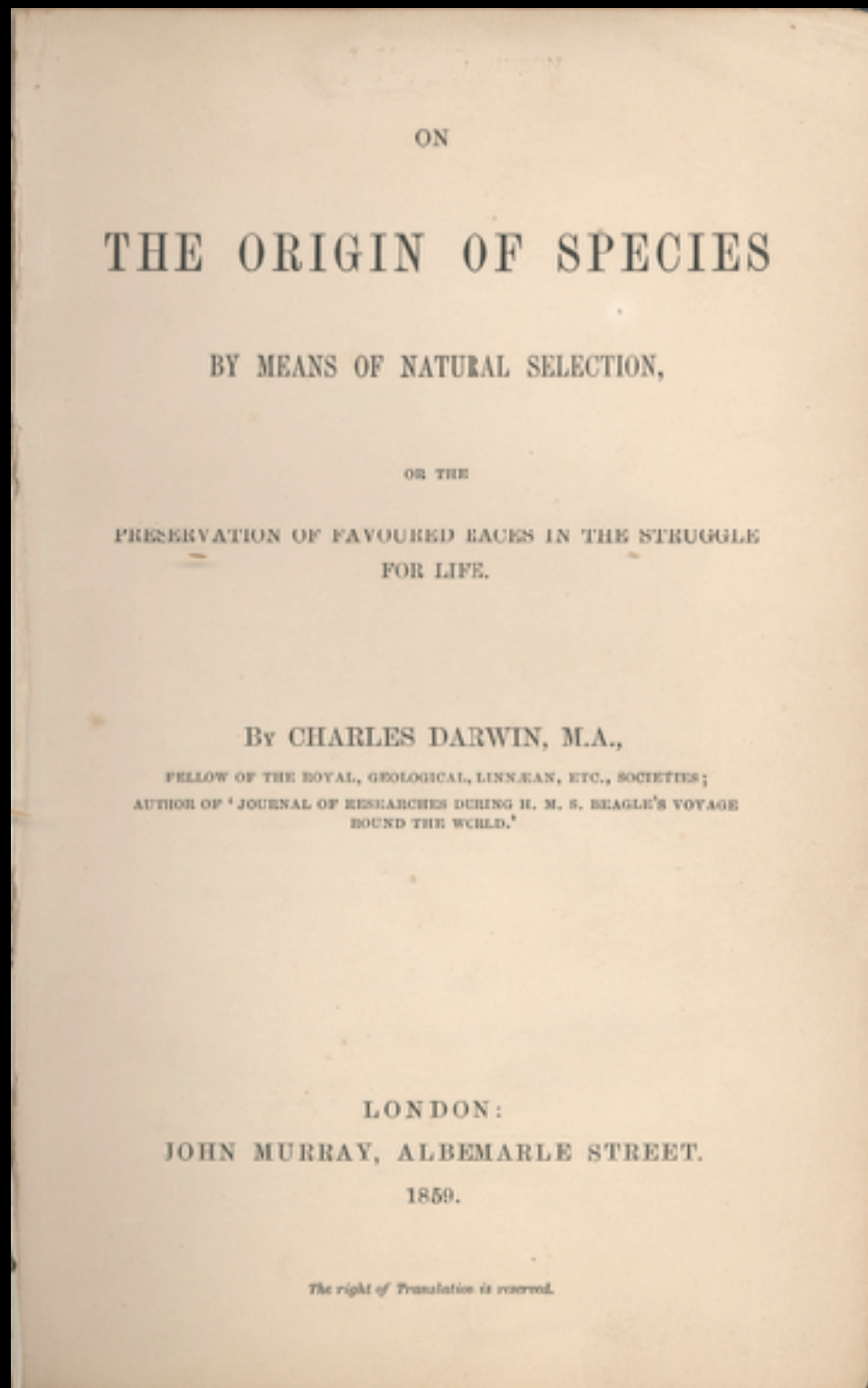
- Species change through evolutionary time.
- Evolutionary change is slow.
- Evolution occurs through adaptation to the environment.
- It generally progresses from the simple to the complex.
- Species are related to one another by common descent

Inheritance of acquired characters

The law of nature by which new individuals receive all that has been acquired in organization during the lifetime of their parents is so true, so striking, so much attested by the facts, that there is no observer who has been unable to convince himself of its reality.

Histoire naturelle des animaux sans vertèbres, 1815

Inheritance of acquired characters



Effects of Use and Disuse.—From the facts alluded to in the first chapter, I think there can be little doubt that use in our domestic animals strengthens and enlarges certain parts, and disuse diminishes them; and that such modifications are inherited.

Inheritance of aquired characters

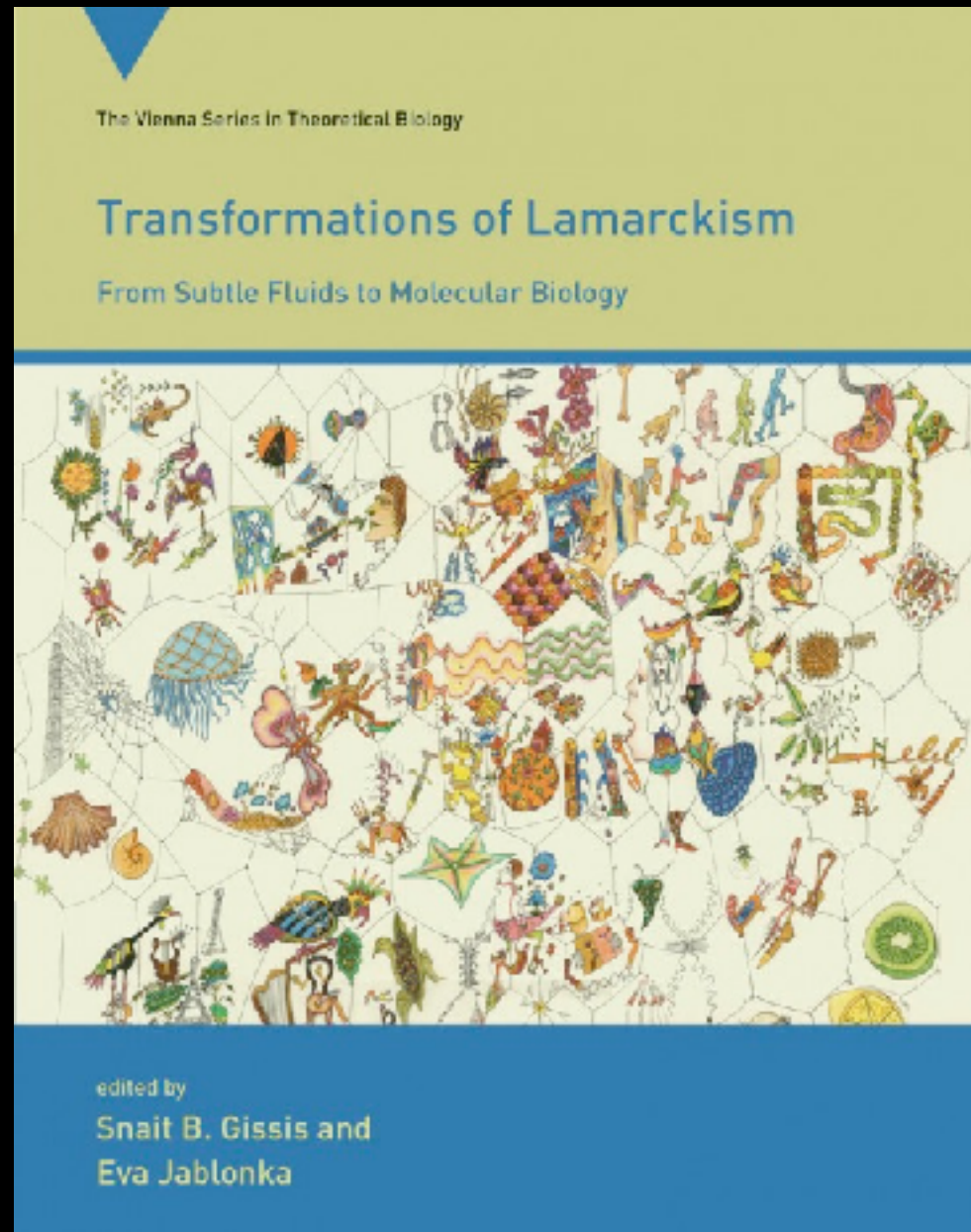
On the Tendency of Species to form Varieties; and on the Perpetuation of Varieties and Species by Natural Means of Selection. By CHARLES DARWIN, Esq., F.R.S., F.L.S., & F.G.S., and ALFRED WALLACE, Esq. Communicated by Sir CHARLES LYELL, F.R.S., F.L.S., and J. D. HOOKER, Esq., M.D., V.P.R.S., F.L.S., &c.

[Read July 1st, 1858.]

London, June 30th, 1858.

Selection acts only by the accumulation of slight or greater variations, caused by external conditions, or by the mere fact that in generation the child is not absolutely similar to its parent.

A revolution in Biology?

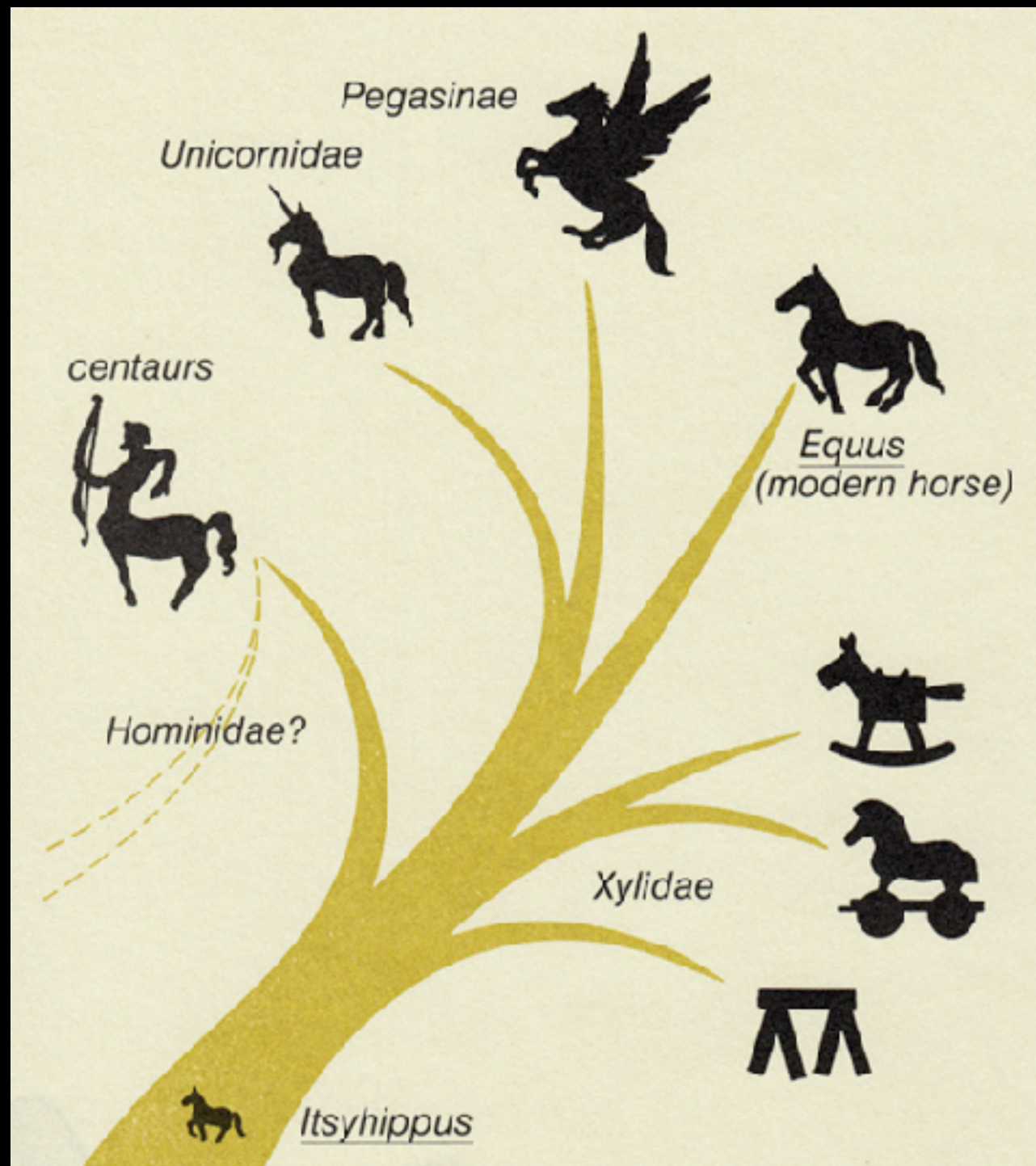


Published 2011

“The problem for us humans is not even the question of the inheritance of acquired characteristics; it is, rather, the question of whether our intentions, as individuals and as groups, have any evolutionary significance, that is, any future significance whatever.”

p.4

Fin.



form *Science made stupid!*, 1985